

A Reproducible TLS 1.3 PKI Lab for PQC Migration

Abstract

Post-quantum TLS experiments need a controlled PKI surface before deployment measurements are meaningful. This manuscript describes a reproducible TLS 1.3 lab for classical, hybrid, and pure-PQC certificate hierarchies, with explicit topologies, certificate material, server configurations, and client probes.

Methodology

The lab fixes the variables that usually blur migration experiments: CA hierarchy, endpoint configuration, supported groups, certificate algorithms, chain shape, and OpenSSL build assumptions. Each scenario records the generated material, negotiated parameters, transcript evidence, and observed failure mode.

Results

The lab gives a stable testbed for comparing PQC migration behavior across certificate hierarchy choices and TLS 1.3 negotiation modes. It is designed for repeatable experiments before wider endpoint measurement.